08_pls_comparison

October 1, 2026

1 Week 4: PLS vs Ridge / LR (LORO)

Supervised compression of AlphaEarth `A00`–`A63` with **Partial Least Squares**, compared to the week-4 balanced pure-cell baselines on the same LORO folds.

AE bands are **not orthogonal** (mean pairwise |corr| ≈ 0.22; effective rank ≈ 10). PLS scores **are** orthogonal by construction — the right place to look for a minimal set of important dimensions.

Data. `datasets/week4_pixel_gt_balanced/pixels.parquet` — 1,751,814 pure RES/NRES cells (1:1 per metro).

Models

Family	What
`baseline_lr` / `baseline_ridge`	StandardScaler + LR / Ridge on all 64 bands
`pls_da_lr`	PLS on binary y → LR on scores (PLS-DA)
`pls_frac`	PLSRegression X → `(frac_res, frac_nres, frac_empty)`
`pls_frac_lr`	PLS on fractions → LR on scores
`pca_lr`	PCA → LR (unsupervised control)

Coarse sweep `n_components` ∈ `[2, 4, 8, 12, 16, 24, 32]`; fine PLS-DA sweep `k=1..20` for the minimum orthogonal set. Backend: **gpu** · 8 GPUs / 80 CPUs · 236.6 s.

```
.venv/bin/python -u scripts/train_week4_pls_loro.py --backend gpu
.venv/bin/python -u scripts/fine_week4_pls_k_sweep.py
.venv/bin/python -u scripts/build_week4_pls_notebook.py
```

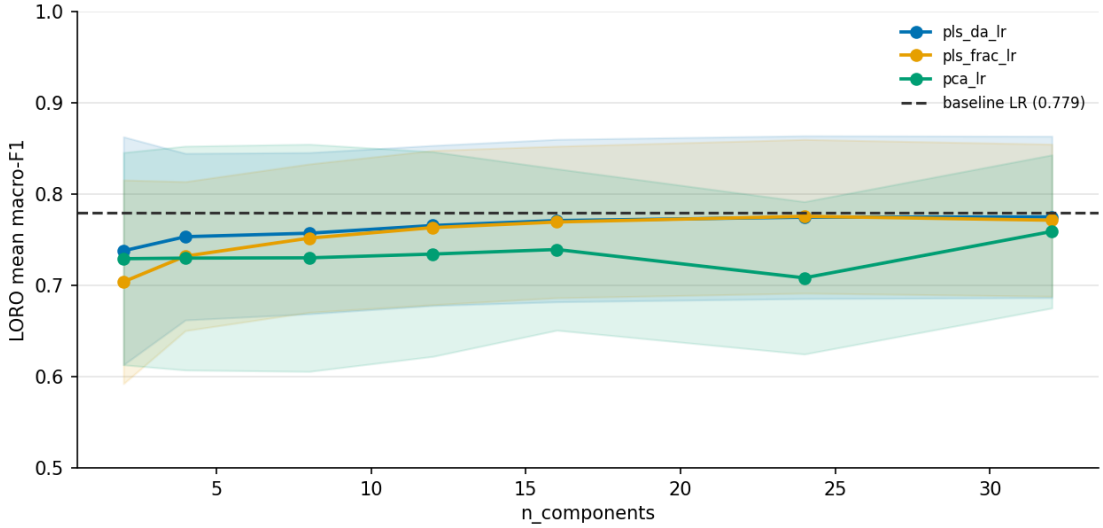
1.1 Headline

Model	LORO macro-F1 / MAE	OOF
Baseline LR (64-D)	F1 0.779 ± 0.085	0.805
PLS-DA → LR (k=32)	F1 0.775 ± 0.088	0.804
PLS frac → LR (k=24)	F1 0.775 ± 0.084	—
PCA → LR (k=32)	F1 0.759 ± 0.084	—
Baseline Ridge	MAE 0.263 ± 0.027	0.254
PLS frac (k=12)	MAE 0.260 ± 0.028	0.253

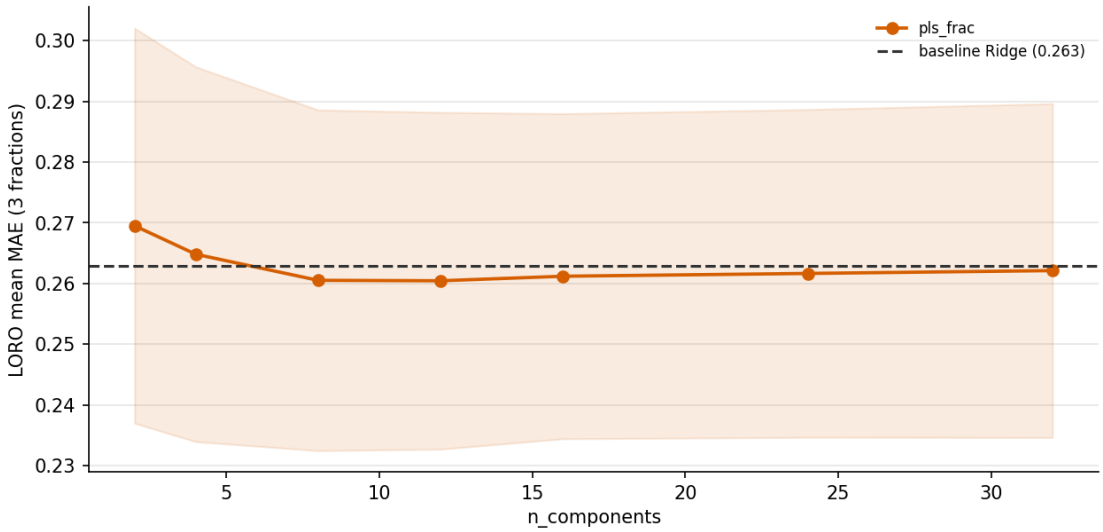
Δ OOF F1 (PLS-DA − baseline) = **-0.0013** · Δ OOF MAE (PLS − Ridge) = **-0.0008**

PLS-DA matches full LR once ~16 components; PCA trails

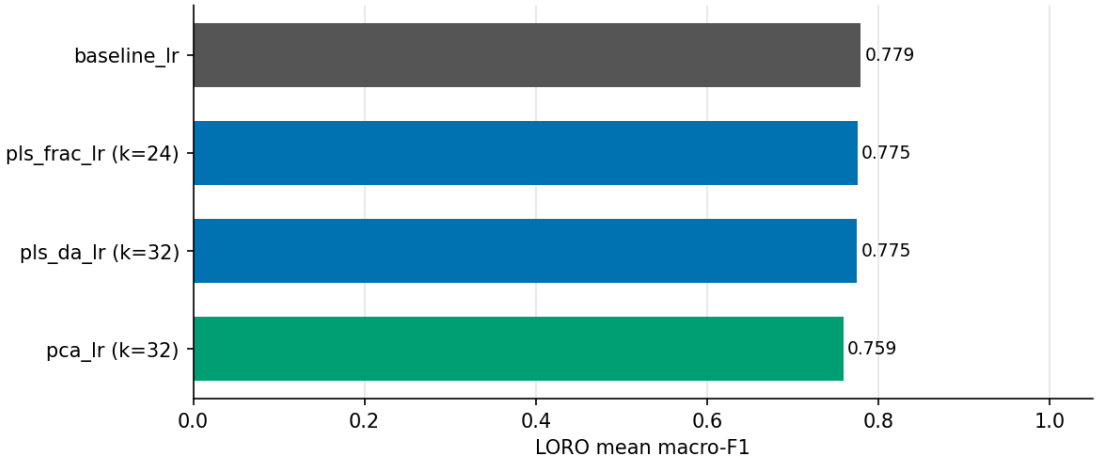
Shaded = ± LORO std across metros. Balanced pure 10 m cells.



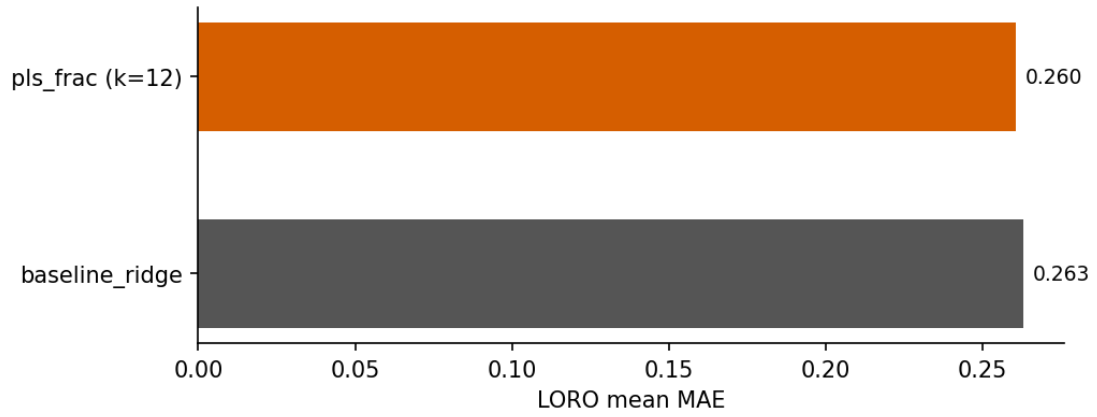
PLS fraction head vs Ridge on all 64 bands

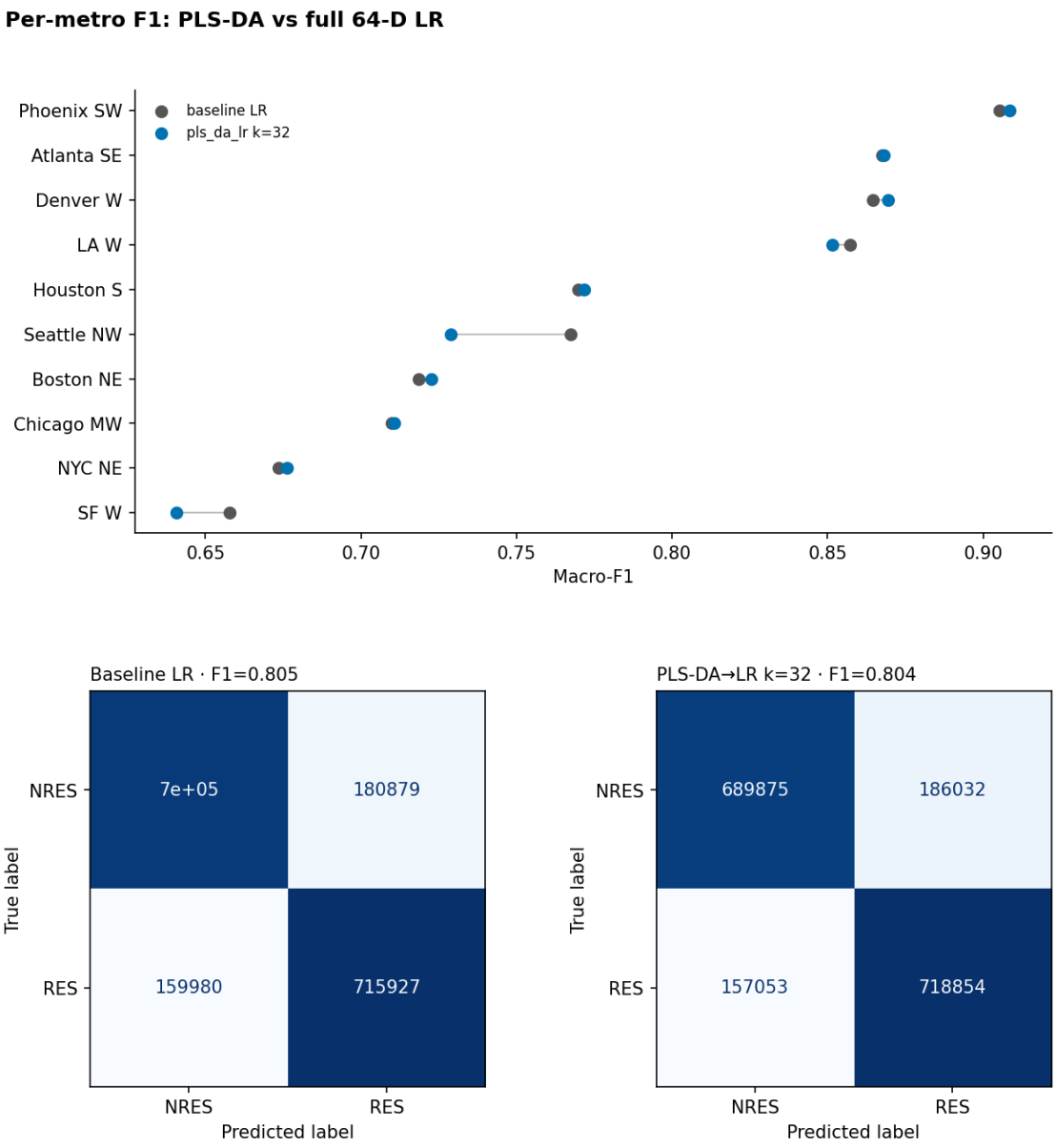


Best-per-family LORO F1: PLS ≈ full LR; PCA weaker



Fraction MAE: PLS vs Ridge





1.2 Do RES / NRES form two clusters in (t_1, t_2)?

Partly — and the hard cities tell you *how* the blobs fail.

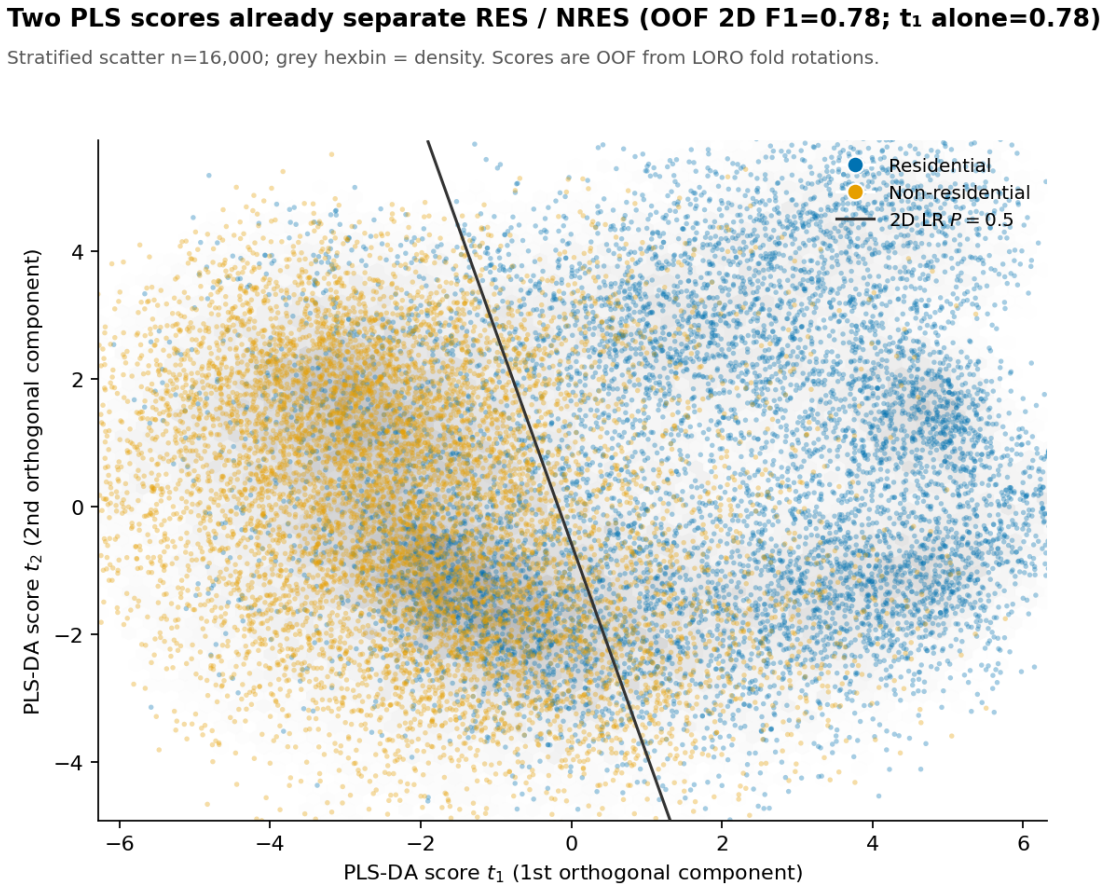
Orange = non-residential, blue = residential. Globally the mass separates along t_1 (NRES left, RES right); the $P = 0.5$ cut is nearly vertical, so t_2 is a thin residual. Marginals: RES peaks near $t_1 \approx +4$, NRES near $t_1 \approx -2.5$, with a real overlap band around $t_1 \in [-1, +2]$.

Probe on pooled OOF scores	macro-F1
1D LR on t_1 only	0.778
2D LR on (t_1, t_2)	0.783

Pooled-OOF F1 is descriptive. Honest LORO at k=2 is ~0.74 — transfer is harder than a global

2D cut.

```
.venv/bin/python -u scripts/plot_week4_pls_t1_t2.py
```



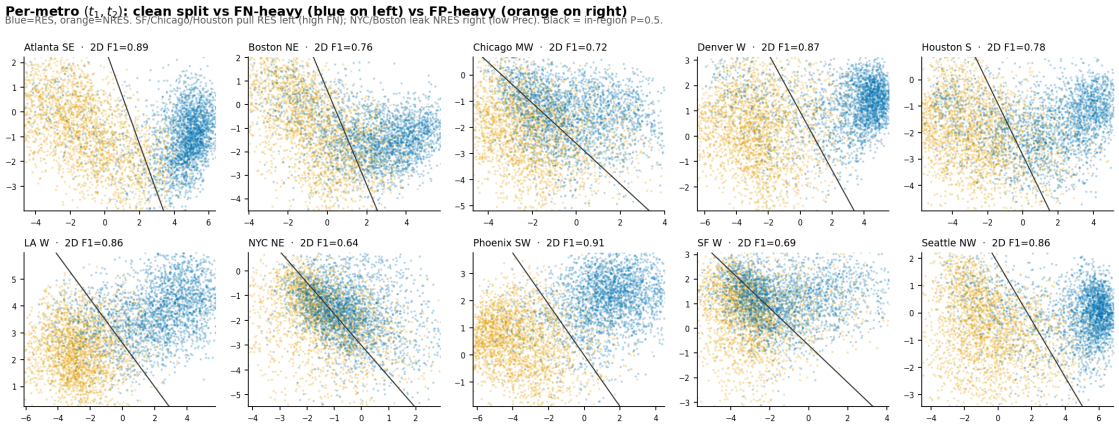
1.2.1 Reading the per-metro panels with precision / recall

What "two clear clusters" looks like in (t_1, t_2) is the same signal the OOF LR metrics quantify. Join in-panel 2D structure with baseline-LR Prec/Recall/FN (RES = positive):

Metro	in-panel 2D F1	mean t_1 RES	mean t_1 NRES	Prec RES	Recall RES	FN rate	What the plot is doing
SF_W	0.688	-1.26	-2.77	0.763	0.490	**0.510**	FN-heavy (RES→NRES)
NYC_NE	0.644	-0.55	-1.34	0.661	0.715	**0.285**	FP-heavy (NRES→RES)
Chicago_MW	0.725	-0.12	-1.83	0.814	0.562	**0.438**	FN-heavy (RES→NRES)
Boston_NE	0.756	2.29	-0.42	0.679	0.847	**0.153**	FP-heavy (NRES→RES)
Seattle_NW	0.864	4.56	-0.46	0.706	0.940	**0.060**	FP-heavy (NRES→RES)
Houston_S	0.784	1.38	-2.45	0.930	0.599	**0.401**	FN-heavy (RES→NRES)
LA_W	0.865	2.07	-2.52	0.812	0.931	**0.069**	clean split
Denver_W	0.866	2.93	-2.48	0.849	0.886	**0.114**	clean split
Atlanta_SE	0.892	4.23	-0.89	0.840	0.909	**0.091**	clean split
Phoenix_SW	0.908	1.16	-3.56	0.945	0.861	**0.139**	clean split

How to read the blobs

1. **Clean two-blob cities** (Phoenix, Atlanta, Denver, LA) — class means on t_1 sit far apart (often $\Delta t_1 \gtrsim 4$). FN rate is low and Prec RES stays high. The black cut sits in a low-density gap; little ink in the wrong lobe.
2. **FN-heavy cities** (SF FN=0.51, Chicago 0.44, Houston 0.40) — a large blue mass still sits on the **left** / **NRES side** of t_1, or the RES cloud is pulled left so many true RES cells fall past the cut. Plot cue: blue points mixed into the orange lobe; high FN, often still decent Prec RES (model is shy about calling RES). Chicago's RES mean t_1 is near **0** — almost no rightward RES lobe.
3. **FP-heavy cities** (NYC, Boston, and Seattle on metrics) — orange bleeds **right** into the RES side. Prec RES drops and/or NRES recall is weak. Plot cue: orange points across the cut into the blue region; FN rate alone can look "ok" (Boston 0.15, Seattle 0.06) while precision is the problem.
4. **NYC** / **SF** / **Chicago** are the visually overlapped panels — not one failure mode. SF/Chicago miss RES; NYC overlaps both ways with low Prec.



Bottom line for the cluster plots: two modes exist along t_1, but hard metros are not "no clusters" — they are **shifted / overlapping modes** whose failure type (FN vs FP) you can see as blue-on-the-left vs orange-on-the-right.

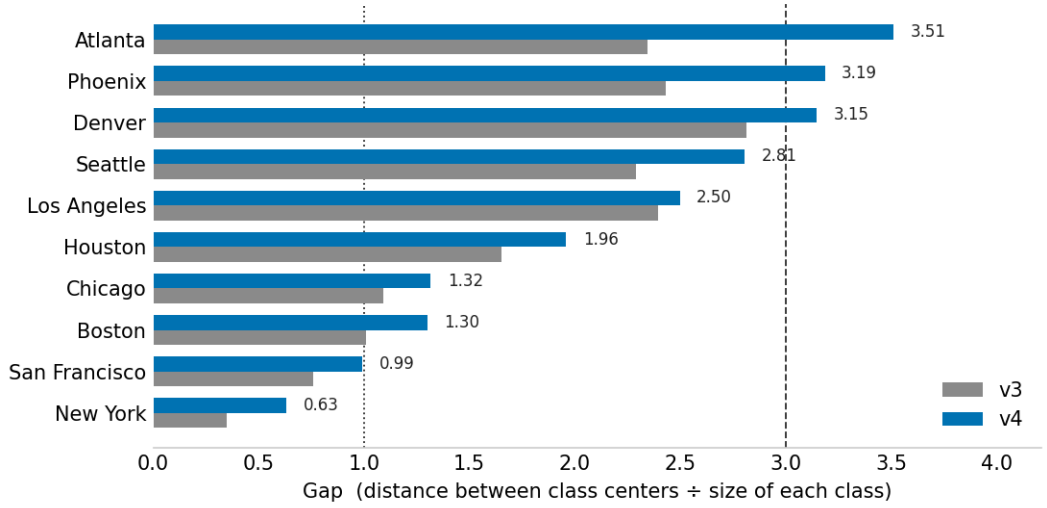
1.3 Held-out 3D clusters after the within-region loss

Each panel is that metro's raw encoder output on the tiles the fold did not train on. The axes are encoder 1, 2, and 3. Blue is residential. Orange is non-residential. The gap is the distance between the two class centers divided by the mean radius. A gap of 3 is a clear separation. A gap below 1 is one cloud. v3 is the previous contrastive run.

Metro	v4 F1	v3 F1	v4 gap	v3 gap
Atlanta	0.898	0.840	3.51	2.34
Phoenix	0.878	0.849	3.19	2.43
Denver	0.862	0.852	3.15	2.81
Seattle	0.824	0.787	2.81	2.29
Los Angeles	0.793	0.842	2.50	2.39
Houston	0.709	0.711	1.96	1.65
Chicago	0.553	0.622	1.32	1.09
Boston	0.693	0.698	1.30	1.01
San Francisco	0.584	0.595	0.99	0.76
New York	0.586	0.552	0.63	0.35

Atlanta, Phoenix, and Denver clear a gap of 3

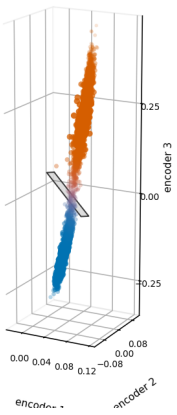
Held-out 100 m tiles. Blue is v4. Grey is v3. Dotted line = 1 (one cloud). Dashed line = 3 (clear gap).

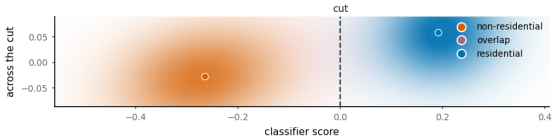


1.3.1 Atlanta

Atlanta: residential and non-residential are apart

Held-out tiles. F1 0.898. Gap 3.51. Color mixes the two centers: blue residential, orange non-residential, mauve where both still reach. Influence falls with distance from a center. The plane and th

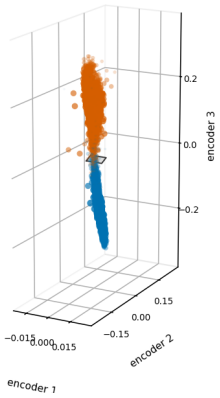


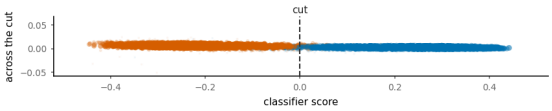


1.3.2 Phoenix

Phoenix: residential and non-residential are apart

Held-out tiles. F1 0.878. Gap 3.19. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the dashed line are the classifier cut.

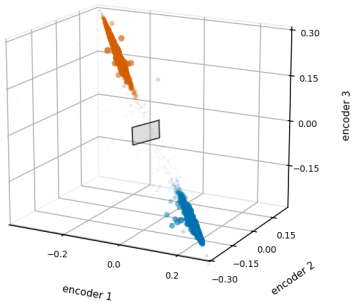


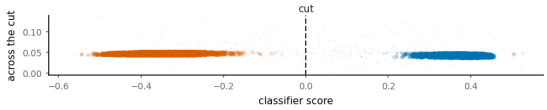


1.3.3 Denver

Denver: residential and non-residential are apart

Held-out tiles. F1 0.862. Gap 3.15. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the dashed line are the classifier cut.

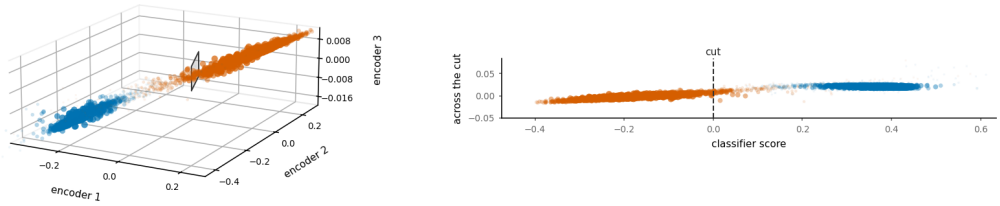




1.3.4 Seattle

Seattle: residential and non-residential are apart

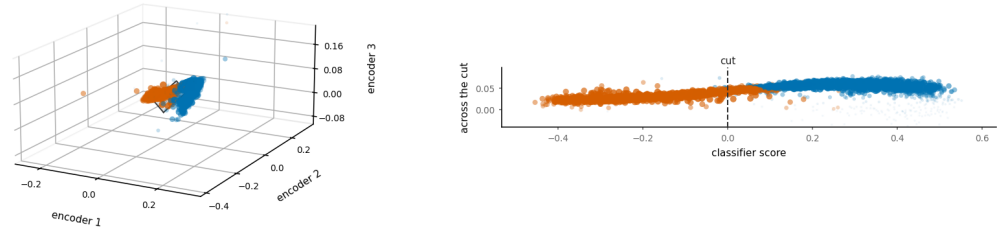
Held-out tiles. F1 0.824. Gap 2.81. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the dashed line are the classifier cut.



1.3.5 Los Angeles

Los Angeles: the two classes touch

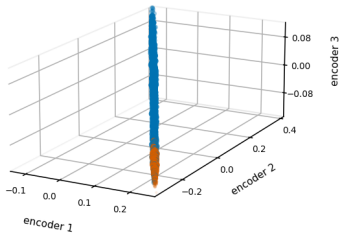
Held-out tiles. F1 0.793. Gap 2.50. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the dashed line are the classifier cut.

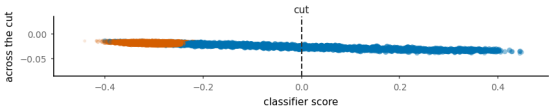


1.3.6 Houston

Houston: the two classes touch

Held-out tiles. F1 0.709. Gap 1.96. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the dashed line are the classifier cut.

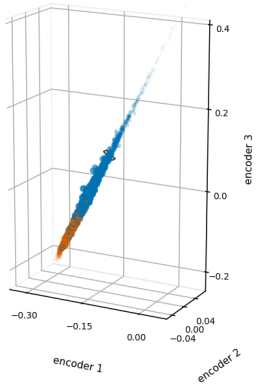


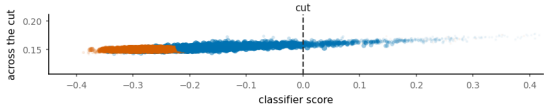


1.3.7 Chicago

Chicago: residential and non-residential are one cloud

Held-out tiles. F1 0.553. Gap 1.32. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the dashed line are the classifier cut.

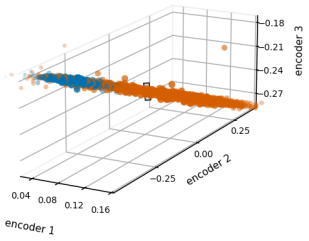


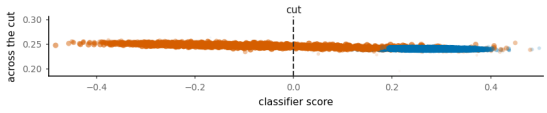


1.3.8 Boston

Boston: residential and non-residential are one cloud

Held-out tiles. F1 0.693. Gap 1.30. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the dashed line are the classifier cut.

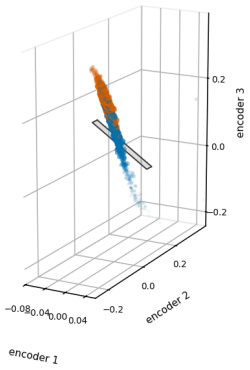


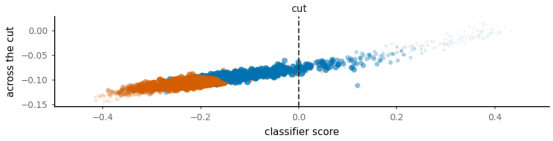


1.3.9 San Francisco

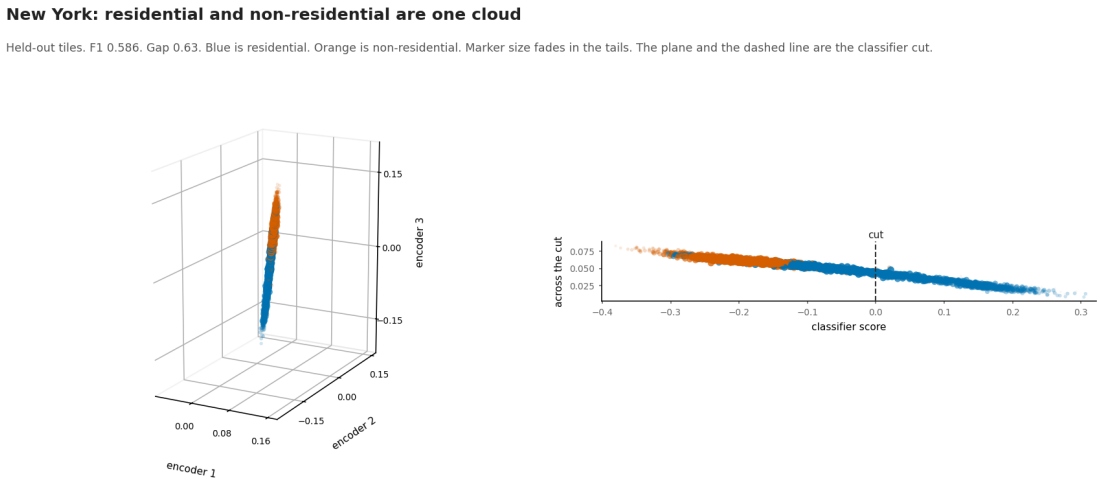
San Francisco: residential and non-residential are one cloud

Held-out tiles. F1 0.584. Gap 0.99. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the dashed line are the classifier cut.





1.3.10 New York



Each fold has its own axes. The coordinates are not pooled across metros.

1.4 Validation clusters while New York is held out

These are the tiles the NYC fold used to decide when to stop. They are a 10% split of the other nine metros, embedded with the checkpoint that had the highest validation gap (epoch 25, mean gap 3.33). The same fold's held-out New York gap is 0.63.

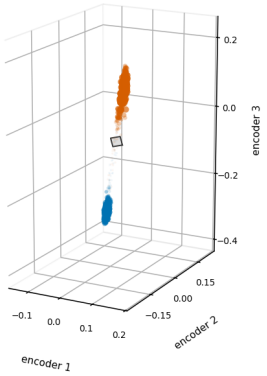
The four widest validation gaps are shown, then San Francisco, the narrowest in this split. Blue is residential. Orange is non-residential.

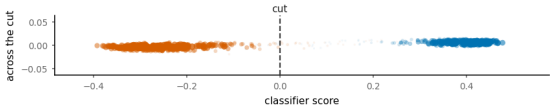
Metro	Validation gap	Tiles
Phoenix	5.44	514
Atlanta	4.98	367
Denver	3.74	743
Los Angeles	3.71	1,068
Seattle	3.59	225
Houston	2.80	637
Chicago	2.01	505
Boston	1.90	550
San Francisco	1.82	111

1.4.1 Phoenix, validation gap 5.44

Phoenix: residential and non-residential are apart

Validation tiles while New York is held out. n=514 (289 non-residential, 225 residential). Gap 5.44. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the da

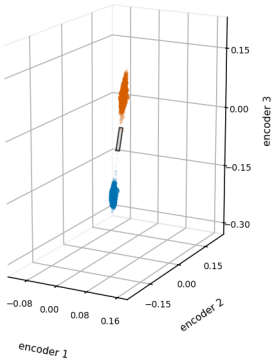


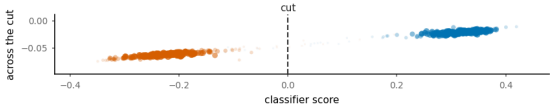


1.4.2 Atlanta, validation gap 4.98

Atlanta: residential and non-residential are apart

Validation tiles while New York is held out. n=367 (180 non-residential, 187 residential). Gap 4.98. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the da

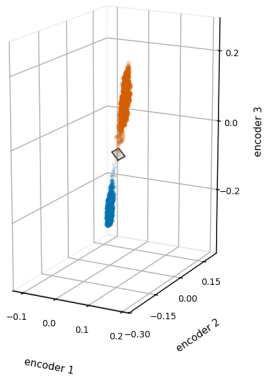


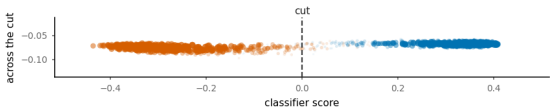


1.4.3 Denver, validation gap 3.74

Denver: residential and non-residential are apart

Validation tiles while New York is held out. n=743 (380 non-residential, 363 residential). Gap 3.74. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the da

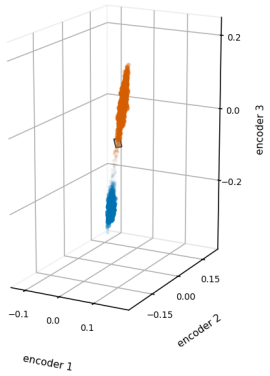


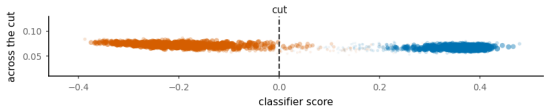


1.4.4 Los Angeles, validation gap 3.71

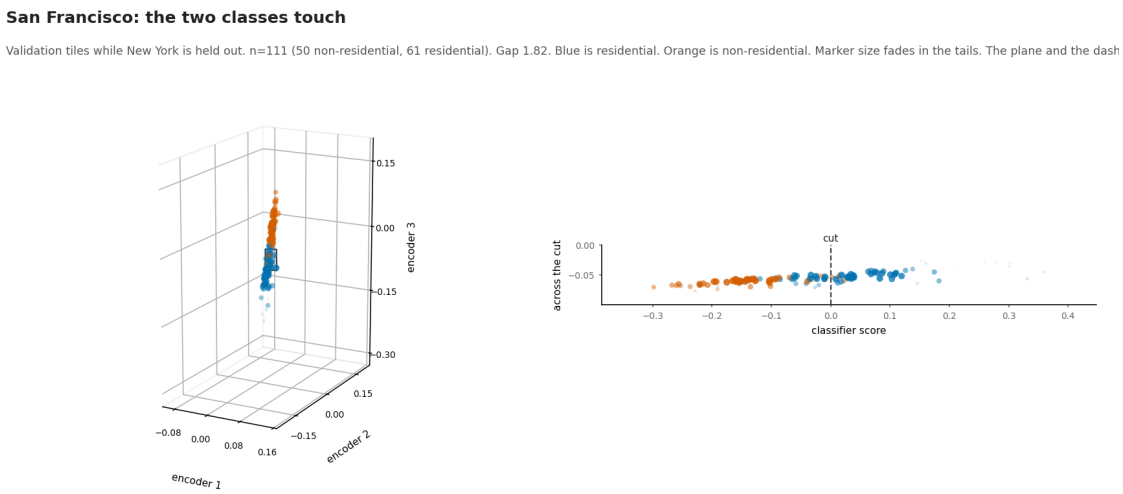
Los Angeles: residential and non-residential are apart

Validation tiles while New York is held out. n=1068 (531 non-residential, 537 residential). Gap 3.71. Blue is residential. Orange is non-residential. Marker size fades in the tails. The plane and the c





1.4.5 San Francisco, validation gap 1.82



The classes sit at opposite ends of a thin line. Phoenix and Atlanta have a visible gap along that line. San Francisco's two ends still touch. This is the training-metro split, not the held-out city.

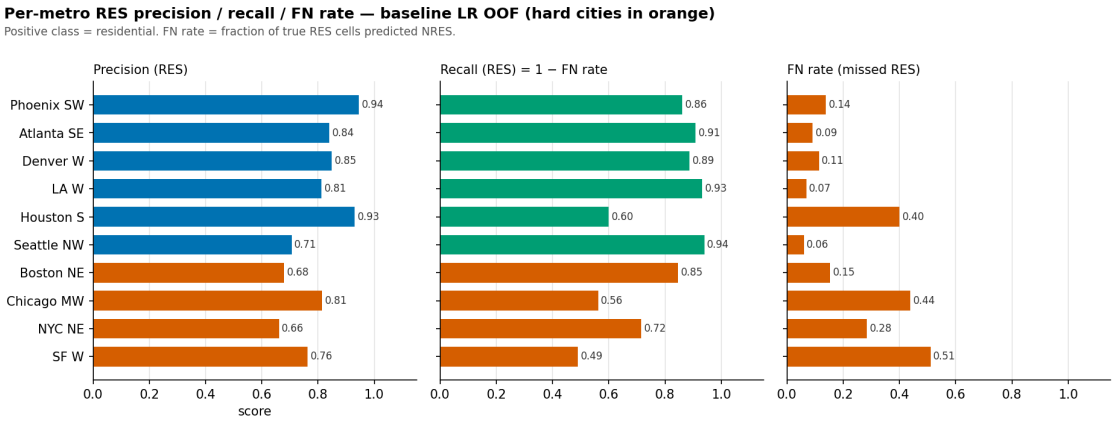
1.5 Per-metro precision / recall / FN rate

Positive class = **residential**. FN rate = missed RES / all true RES = $1 - \text{recall}_{\text{res}}$. Same numbers used to annotate the cluster panels above.

Baseline LR OOF (LORO), sorted by RES FN rate:

Metro	Prec RES	Recall RES	FN rate RES	Prec NRES	Recall NRES	macro-F1	
SF_W	0.763	0.490	**0.510**	0.624	0.848	0.658	← hard
Chicago_MW	0.814	0.562	**0.438**	0.666	0.872	0.710	← hard
Houston_S	0.930	0.599	**0.401**	0.704	0.955	0.770	
NYC_NE	0.661	0.715	**0.285**	0.690	0.633	0.673	← hard
Boston_NE	0.679	0.847	**0.153**	0.797	0.599	0.719	← hard
Phoenix_SW	0.945	0.861	**0.139**	0.873	0.950	0.905	
Denver_W	0.849	0.886	**0.114**	0.881	0.843	0.864	
Atlanta_SE	0.840	0.909	**0.091**	0.901	0.827	0.868	
LA_W	0.812	0.931	**0.069**	0.920	0.785	0.857	
Seattle_NW	0.706	0.940	**0.060**	0.910	0.608	0.768	

Group	mean FN rate RES	mean Recall RES	mean Prec RES	mean macro-F1
Hard (NYC, SF, Chicago, Boston)	0.347	0.653	0.729	0.690
Other	0.145	0.855	0.847	0.839




```
[ ]: pr = pd.read_csv(OUT / "per_region_precision_recall.csv")
pr.loc[pr.model == "baseline_lr", [
    "region", "precision_res", "recall_res", "fn_rate_res",
    "precision_nres", "recall_nres", "macro_f1", "fn", "fp"
]].sort_values("fn_rate_res", ascending=False)
```

1.6 Equation: first 10 PLS-DA components → P(residential)

Fold-averaged PLS-DA → logistic (LORO). Score means ≈ 0, so $z_k \approx t_k / s_k$.

$$\tilde{x}_j = \frac{A_j - \mu_j}{\sigma_j}, \qquad t_k = \mathbf{r}_k^\top \tilde{\mathbf{x}}, \quad k = 1, \ldots, 10$$

$$\text{logit} = +0.0952 + 0.7071 t_1 + 0.3750 t_2 + 0.2715 t_3 + 0.2357 t_4 + 0.3275 t_5 + 0.2488 t_6 + 0.1374 t_7 + 0.0863 t_8 + 0.1481 t_9 + 0.1426$$

$$P(\text{RES} \mid \mathbf{x}) = \sigma(\text{logit}) = \frac{1}{1 + e^{-\text{logit}}}$$

Numerically (same thing):

```
logit = +0.0952 +0.7071*t1 +0.3750*t2 +0.2715*t3 +0.2357*t4 +0.3275*t5 +0.2488*t6 +0.1374*t7 +0.
P_res = 1 / (1 + exp(-logit))
```

OOF macro-F1 (fold-wise $k = 10$): **0.793**.

Full $\mathbf{r}_k$ (64×10), $\boldsymbol{\mu}$, $\boldsymbol{\sigma}$, and compact $\mathbf{c}$ on scaled AE: `outputs/week4_pls_loro/pls_da_k10_*.{json,csv,md}`.

component	β_k (on z_k)	s_k	coef on t_k (β_k / s_k)
t1	2.2401	3.1681	0.7071
t2	0.7896	2.1057	0.3750
t3	0.5723	2.1076	0.2715
t4	0.3740	1.5867	0.2357

component	β_k (on z_k)	s_k	coef on t_k (β_k/s_k)
t5	0.4173	1.2740	0.3275
t6	0.3399	1.3660	0.2488
t7	0.2117	1.5413	0.1374
t8	0.1349	1.5641	0.0863
t9	0.1560	1.0533	0.1481
t10	0.1453	1.0188	0.1426

[ ]:
```
betas = pd.read_csv(OUT / "pls_da_k10_lr_betas.csv")
R = pd.read_csv(OUT / "pls_da_k10_loadings_R.csv", index_col=0)
c = pd.read_csv(OUT / "pls_da_k10_compact_c.csv")
print(betas.to_string(index=False))
print("\nTop |c| on scaled AE:")
print(c.reindex(c.c.abs().sort_values(ascending=False).index).head(12).
 ↪to_string(index=False))
```

1.7 Load outputs

[ ]:
```
from pathlib import Path
import json
import pandas as pd

ROOT = Path("../..").resolve()
OUT = ROOT / "outputs" / "week4_pls_loro"
summary = json.loads((OUT / "summary.json").read_text())
sweep = pd.read_csv(OUT / "sweep_summary.csv")
metrics = pd.read_csv(OUT / "loro_metrics.csv")
fine = json.loads((OUT / "fine_k_sweep.json").read_text())
fine_df = pd.read_csv(OUT / "fine_k_sweep.csv")
print("baseline LORO F1:", round(fine["baseline_loro_f1"], 4))
print("min-k criteria:", json.dumps(fine["min_k_criteria"], indent=2))
print("cov cumulants: 90%@k=%s  95%@k=%s  99%@k=%s" % (
    fine["cov_cum_k90"], fine["cov_cum_k95"], fine["cov_cum_k99"]))
```

1.8 Minimum orthogonal dimensions (fine PLS-DA sweep)

PLS-DA scores are **orthogonal**. Fine LORO sweep over `k = 1..20` on the same balanced pure cells (8 GPUs).

```
.venv/bin/python -u scripts/fine_week4_pls_k_sweep.py
```

1.8.1 Covariance with (y) (signal concentration)

Almost all supervised covariance sits in the first few scores:

Cumulative $\text{cov}^2(t, y)$	k
~90%	**1**
~95%	**2**
~99%	**5**

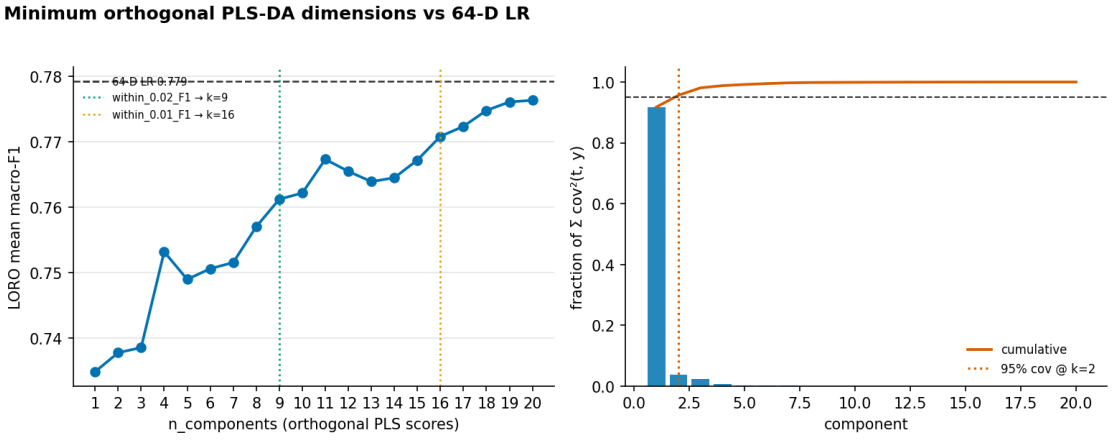
1.8.2 Classifier parity vs 64-D LR (LORO F1 = 0.7792)

Criterion	Min k	F1	gap
within 0.02 F1	**9**	0.7612	0.0180
within 0.01 F1	**16**	0.7708	0.0084
within 2% relative	**11**	0.7673	0.0119
within 1% relative	**17**	0.7723	0.0069

1.8.3 Selected k on the fine curve

k	LORO F1	gap to 64-D	rel gap
1	0.735	0.0444	0.0569
2	0.738	0.0415	0.0532
5	0.749	0.0302	0.0388
9	0.761	0.0180	0.0231
11	0.767	0.0119	0.0153
12	0.765	0.0137	0.0176
16	0.771	0.0084	0.0108
17	0.772	0.0069	0.0088
20	0.776	0.0029	0.0037
64	0.779	0.0000	0.0000

Recommendation - Absolute minimum for the residential *signal*: **k = 5** (99% of cov with (y)). - Best compression that still races LR: **k = 9–12**. - Near-parity with full 64-D LR: **k = 16**.



[ ]:
```
fine_df
```

1.8.4 Binary heads only (coarse sweep)

[ ]:
```
bin_models = ["baseline_lr", "pls_da_lr", "pls_frac_lr", "pca_lr"]
sweep.loc[sweep.model.isin(bin_models), [
    "model", "n_components", "loro_macro_f1_mean", "loro_macro_f1_std"
]].sort_values(["model", "n_components"])
```

1.8.5 Fraction heads (coarse sweep)

[ ]:
```
frac_models = ["baseline_ridge", "pls_frac"]
sweep.loc[sweep.model.isin(frac_models), [
    "model", "n_components", "loro_mae_mean", "loro_mae_std"
]].sort_values(["model", "n_components"])
```

1.9 Takeaways

1. **AE bands are collinear** (not orthogonal; effective rank ~10). PLS scores *are* orthogonal — use those as the compressed features.
2. **PLS ≈ full LR / Ridge** at enough components (Δ OOF F1 ≈ −0.001; fraction MAE slightly better at k=12). PCA at the same width lags → the gain is supervised, not just dim reduction.
3. **Minimum orthogonal set**
   - signal (99% cov with (y)): **k = 5**
   - race LR (within 0.02 F1): **k = 9–12**
   - near-parity (within 0.01 F1): **k = 16**
4. Artifacts under `outputs/week4_pls_loro/`: OOF preds, best-k PLS scores (`oof_pls_da_scores_k32.npy`), fold checkpoints (loadings/rotations), coarse + fine sweeps, figures.

[ ]:
```
print("paths:", json.dumps(summary.get("paths", {}), indent=2))
print("backend:", summary.get("backend"), "gpus:", summary.get("n_gpus"), "cpus:
 ↪", summary.get("n_cpus"))
print("elapsed_s:", summary.get("elapsed_s"))
if (OUT / "fine_k_sweep.json").exists():
    print("fine_k_sweep:", OUT / "fine_k_sweep.json")
    print("fine_k_figure:", OUT / "figures" / "fine_k_minimum.png")
```